

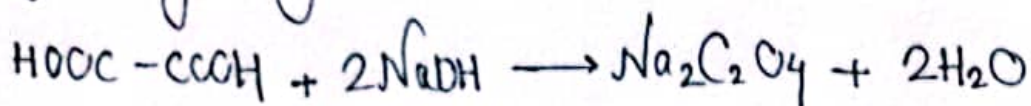
Objective:

To find the normality of a NaOH solution using a standard solution of oxalic acid.

Theory:

Titration is the slow addition of one solution of a known concentration (called titrant) to a known volume of another solution of unknown concentration until the reaction reaches the point of neutralization, which is often indicated by a colour change.

Oxalic acid reacts with sodium hydroxide in the following way:



And in order to conduct the titration, phenolphthalein is used as an indicator. It turns the colour from pink to colourless by the method of titration.

The normality of oxalic acid solution used was 0.1 N.

Apparatus:

1. Conical flask.
2. Pipette
3. Burette.
4. Funnel.
5. Stand and clamp.

Chemicals used :

1. NaOH (acts as the base)
2. $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (acts as the acid)
3. Phenolphthalein (indicator).

Procedure:

A. Preparation of Oxalic Acid solution:

We transfer 0.63 gm of pure oxalic acid ($\text{HOOC}-\text{COOH} \cdot 2\text{H}_2\text{O}$) in a 100 mL. measuring flask and dissolve it in distilled water and it is marked at a certain level. The normality of the prepared solution is measured as follows:

$$N_{\text{acid}} = \frac{\text{Weight taken in gm} \times 0.1 N}{0.63 \text{ gm}}$$

B. Finding out the concentration using titration method:

We take 10 mL of NaOH solution in a conical flask by the help of a pipette. This is diluted to about 50 mL. Next, 1-2 drops of phenolphthalein is added to the solution as an indicator. Using a burette, standard oxalic acid solution is added slightly drop by drop. While oxalic acid is being added, the flask is shaken simultaneously. The colour change of the solution is noticed carefully and as soon as the pink colour of the solution just disappears, we stop adding the oxalic acid. At this state, the burette reading is noted down. The reading should be taken at the lower meniscus of the liquid. The difference between the initial and final reading of the burette gives the volume of added acid solution. The process has to be repeated at least three times to minimize error.

be agreed within ± 0.1 mL. Using the following relation, the normality of NaOH solution is determined:

$$V_{\text{acid}} \times S_{\text{acid}} = V_{\text{base}} \times S_{\text{base}}$$

Experimental Data:

No. of observations	Volume of NaOH solution taken (mL)	Burette Reading (mL)			Average volume of oxalic soln required (mL)
		Initial	Final	Difference	
1	10			10.52	10.4567.
2				10.45	
3				10.40	

Calculation:

$$V_{\text{oxalic acid}} \times S_{\text{oxalic acid}} = V_{\text{NaOH}} \times S_{\text{NaOH}}$$

$$\Rightarrow 10.4567 \times 0.1 = 10 \times S_{\text{NaOH}}$$

$$\Rightarrow S_{\text{NaOH}} = \frac{10.4567 \times 0.1}{10}$$

$$\therefore S_{\text{NaOH}} = 0.104567 \text{ N.}$$

Results:
Thus, the concentration of the unknown NaOH solution is found to be:

$$S = 0.104567 \text{ N.}$$

Error Calculation:

The known value of concentration = 0.105 N

The practical value which is obtained from experiment = 0.104567 N.

$$\therefore \% \text{ error} = \frac{0.105 - 0.104567}{0.105} \times 100\% \\ = 0.41\%$$

Discussion:

1. All equipments need to be washed thoroughly with distilled water before starting the experiment.
2. While we realize that the endpoint is near, we need to carefully add the oxalic acid drop by drop into the volumetric flask.

3. Funnel needs to be removed from the burette during titration.
4. When the colour of the indicator turns colourless, at that instant, the titration should be stopped.
5. The volumetric flask needs to be placed above a clean white paper to determine whether the solution became completely colourless or not.
6. Phenolphthalein is used as an indicator because it changes colour in the pH range 8.3-10.